

Object / Blob Storage Basics

Phase 1 · Fundamentals · Estimated time: 1 hour

1. Concept From First Principles

English:

Not everything belongs in a database. A profile photo, a PDF invoice, a video -- these are large, unstructured files, and stuffing them into a database or local disk causes real problems at scale: a file saved to one server's disk isn't visible to the other 49 servers behind your load balancer.

Hindi:

हर चीज़ database में नहीं रखी जाती। एक profile photo, एक PDF invoice, एक video -- यह बड़े, unstructured files होते हैं, और इन्हें database या local disk में ठूसने से बड़े स्तर पर असली समस्याएं आती हैं: एक server की disk पर save की गई file आपके load balancer के पीछे बाकी 49 servers को देखिती ही नहीं।

Important Words:

- **Stuffing (figurative)** – ठूसना / जबरदस्ती भरना

English:

Object storage (AWS S3, Google Cloud Storage, Azure Blob Storage) solves this: it's a separate, purpose-built service for storing and retrieving files over a simple HTTP API, completely decoupled from your application servers and database.

Hindi:

Object storage (AWS S3, Google Cloud Storage, Azure Blob Storage) इसका हल है: यह files को store और retrieve करने के लिए एक अलग, खास तौर पर बना हुआ service है, जो एक साधारण HTTP API पर काम करता है और आपके application servers व database से पूरी तरह अलग (decoupled) होता है।

Important Words:

- **Purpose-built** – खास मकसद के लिए बना
- **Decoupled** – अलग किया हुआ

Backend engineer analogy

English:

If you've ever used Laravel's `Storage::disk('s3')->put(...)` instead of saving to `local storage/app/public`, you've already made this exact HLD decision -- local disk doesn't survive a server restart, isn't shared across app servers, and doesn't scale.

Hindi:

अगर आपने कभी `local storage/app/public` में save करने की बजाय Laravel का `Storage::disk('s3')->put(...)` use किया है, तो आप पहले से यही HLD decision ले चुके हैं -- local disk server restart के बाद नहीं बचती, यह app servers के बीच shared नहीं होती, और scale नहीं करती।

Object storage vs. file storage vs. block storage

English:

One more property that matters: object storage is typically durable by design -- providers like S3 replicate every object across multiple physical facilities automatically, often advertising eleven-nines durability.

Hindi:

एक और ज़रूरी बात: object storage अक्सर design से ही durable होता है -- S3 जैसे providers हर object को अपने आप कई physical facilities में replicate कर देते हैं, अक्सर eleven-nines जैसी बहुत ऊंची durability का दावा करते हुए।

Important Words:

- **Durable** – टिकाऊ / स्थायी
- **Facility** – सुविधा केंद्र / डेटा सेंटर

English:

- Object storage: flat structure, each file is an 'object' with a unique key and metadata, accessed over HTTP.
- File storage: traditional hierarchical folders/directories, good for POSIX-style file operations.
- Block storage: raw storage volumes attached to a single server, fast but tied to one machine.

Hindi:

- Object storage: flat structure, हर file एक unique key और metadata वाला 'object' होता है, HTTP से access होता है।
- File storage: पारंपरिक hierarchical folders/directories, POSIX-style file operations के लिए अच्छा।
- Block storage: एक ही server से जुड़े raw storage volumes, तेज़ लेकिन एक machine से बंधे हुए।

Important Words:

- **POSIX** – पॉसक्स (एक standard)

English:

For large uploads, well-designed systems don't route the file's bytes through the application server at all. Instead, the app server generates a short-lived, permission-scoped pre-signed URL and hands it to the client, which then uploads directly to object storage.

Hindi:

बड़ी files के upload के लिए, अच्छी तरह design किए गए systems file के bytes को application server से बिल्कुल भी नहीं गुज़ारते। इसके बजाय, app server एक थोड़े समय के लिए valid, permission-सीमति pre-signed URL बनाता है और client को दे देता है, जो फिर सीधे object storage में upload करता है।

Important Words:

- **Short-lived** – थोड़े समय के लिए मान्य

2. Real-World Example / Case Study

English:

When a Zomato/Swiggy-style restaurant partner uploads menu photos, those files go to object storage, not the database. The database only stores a reference -- a URL or object key -- while the actual bytes live in object storage.

Hindi:

जब कोई Zomato/Swiggy जैसा restaurant partner menu photos upload करता है, तो वह files object storage में जाती हैं, database में नहीं। Database सिर्फ एक reference रखता है -- एक URL या object key -- जबकि असली bytes object storage में रहते हैं।

English:

Razorpay and PhonePe store transaction receipts and KYC documents in object storage with strict access controls and encryption at rest.

Hindi:

Razorpay और PhonePe transaction receipts और KYC documents को object storage में सख्त access controls और encryption at rest के साथ रखते हैं।

Important Words:

- **Encryption at rest** – स्थिर डेटा की एन्क्रिप्शन

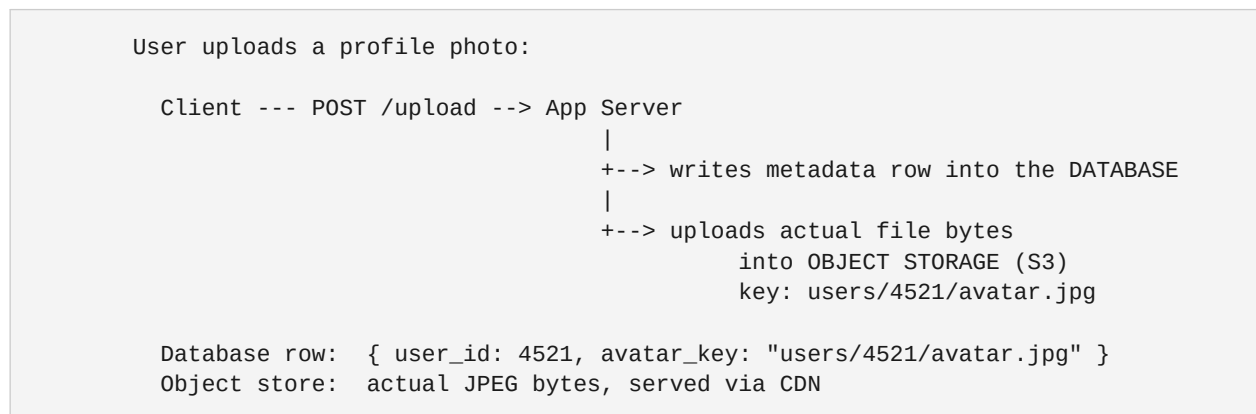
English:

A third example: Talabat's menu-photo pipeline likely uses pre-signed upload URLs for restaurant partners, followed by an asynchronous image-processing step triggered by an event once the upload lands.

Hindi:

एक तीसरा उदाहरण: Talabat की menu-photo pipeline शायद restaurant partners के लिए pre-signed upload URLs use करती है, और upload होते ही एक event से asynchronous image-processing step शुरू हो जाता है।

3. Diagram (English labels kept as-is)



4. Trade-offs Table (Reference Table – English)

Storage type	Best for	Weak for
Object storage (S3-style)	Images, videos, documents, backups	Frequent small in-place edits
File storage (NFS-style)	Shared app config, legacy POSIX apps	Massive horizontal scale
Block storage (EBS-style)	Database disks, single-server storage	Sharing across many servers
Database BLOB column	Tiny files tied to a row's lifecycle	Anything large or frequently accessed

5. Common Interview Questions

Q1: "Where would you store user-uploaded profile photos, and why not the database?"

English:

Model approach: Recommend object storage, explain that storing large binary files in a relational DB bloats table size and slows backups, then describe storing a reference in the DB while the file lives in S3.

Hindi:

Model approach: object storage सुझाएं, समझाएं कि बड़ी binary files को relational DB में रखने से table size बढ़ जाता है और backups धीमे हो जाते हैं, फिर बताएं कि DB में सिर्फ एक reference रखें जबकि file S3 में रहे।

Important Words:

- **Bloat (verb)** – फूल जाना / बढ़ जाना

Q2: "How would you design storage for a video-upload feature?"

English:

Model approach: Object storage for raw and processed video files, a database for video metadata, and mention video processing typically happens asynchronously via a queue after upload.

Hindi:

Model approach: raw और processed video files के लिए object storage, video metadata के लिए एक database, और यह बताएं कि video processing आमतौर पर upload के बाद queue के जरूरी asynchronously होती है।

Q3: "Why route a large file upload directly to object storage instead of through your app server?"

English:

Model approach: Streaming large files through the app server wastes bandwidth, memory, and connection slots. A pre-signed URL lets the app server just authorize the transfer.

Hindi:

Model approach: बड़ी files को app server से गुज़ारना bandwidth, memory, और connection slots बर्बाद करता है। एक pre-signed URL से app server सिर्फ transfer को authorize करता है।

Important Words:

- **Authorize** – अनुमति देना

6. Practice Exercises

English:

- For a food delivery app, list 3 things in object storage and 3 in a relational database, with reasons.
- Explain why saving an uploaded file to an app server's local disk breaks with more than one app server.
- Sketch the data flow from this chapter for a PDF invoice generation feature instead of a photo upload.

Hindi:

- एक food delivery app के लिए 3 चीज़ें object storage में और 3 relational database में बताएं, वजह सहित।

- समझाएं कि app server की local disk पर file save करना एक से ज़्यादा app servers होने पर क्यों टूट जाता है।
- इस chapter का data flow diagram photo upload की बजाय PDF invoice generation के लिए बनाएं।

7. Curated YouTube Videos (Reference – English)

Language	Video & Channel	Why it helps
English	"AWS S3 Explained Object Storage Service"	Direct, focused explanation.
English	"Amazon S3: Overview and Tutorial"	Practical overview of real-world usage.
Hindi	"Amazon S3 Tutorial for Beginners in Hindi" — Great Learning	Hindi walkthrough of S3 basics.
Hindi	"Complete System Design Roadmap 2025" — CodeHelp	Places object storage in the broader roadmap.

8. Key Takeaways

English:

- Large, unstructured files belong in object storage, not the database.
- Object storage is accessed over HTTP, independent of your app servers.
- The database should store only a reference to the file, not the file's bytes.
- Local disk storage on an app server breaks with more than one server instance.
- Object storage is typically fronted by a CDN for fast, cheap delivery.
- Block storage and file storage are different tools for different jobs.

Hindi:

- बड़ी, unstructured files object storage में रहनी चाहिए, database में नहीं।
- Object storage HTTP पर access होता है, आपके app servers से अलग।
- Database को सिर्फ file का reference रखना चाहिए, उसके bytes नहीं।
- App server की local disk एक से ज़्यादा server instances होने पर काम नहीं करती।
- Object storage आमतौर पर तेज़, सस्ती delivery के लिए CDN के पीछे रखा जाता है।
- Block storage और file storage अलग-अलग कामों के लिए अलग-अलग tools हैं।

General Words Glossary

A consolidated list of the important English words used throughout this chapter, with Hindi meanings and simple explanations — useful for revising vocabulary after finishing the chapter.

English Word	Hindi Meaning	Simple Explanation
Authorize	अनुमति देना	To give permission for an action
Bloat (verb)	फूल जाना / बढ़ जाना	To become unnecessarily large
Decoupled	अलग किया हुआ	Not tightly connected to another part
Durable	टकिऊ / स्थायी	Able to survive over time without loss
Encryption at rest	स्थिर डेटा की एन्क्रिप्शन	Encrypting data while it's stored, not just in transit
Facility	सुविधा केंद्र / डेटा सेंटर	A physical location housing servers
POSIX	पॉसिक्स (एक standard)	A standard way operating systems handle files
Purpose-built	खास मकसद के लिए बना	Designed specifically for one job
Short-lived	थोड़े समय के लिए मान्य	Valid only for a limited time
Stuffing (figurative)	ढूसना / जबरदस्ती भरना	Forcing something in without proper design